Project Report

HR Analytics — Employee Attrition Analysis

IBM HR Dataset | Microsoft Excel | Sowmya Sanikommu | 2026

Executive Summary

This report presents a data-driven analysis of employee attrition across 1,470 IBM employee records. The overall attrition rate stands at **16.12%** — exceeding the industry benchmark of 13.2% — indicating a systemic retention challenge rather than isolated incidents.

Analysis reveals that attrition is concentrated in the **Sales department** (20.63% rate), driven heavily by **Sales Representatives** (39.76% attrition), and strongly correlated with **overtime work** (30.53% attrition vs. 10.44% for non-overtime employees). Employees who leave are, on average, **4 years younger** and earn **₹2,046/month less** than those who stay, pointing to an early-career compensation and engagement gap.

Three strategic recommendations are presented: targeted retention programs for high-risk roles, an immediate review of overtime and workload policies, and a strengthened early-career value proposition through compensation benchmarking and structured growth paths.

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1. Introduction

1.1 Business Context

Employee attrition — the rate at which employees voluntarily leave an organization — represents one of the most significant and measurable risks in human capital management. Beyond the direct costs of recruitment and training, attrition disrupts team productivity, erodes institutional knowledge, and negatively impacts employee morale across remaining staff.

Industry research consistently estimates the cost of replacing a single employee at **50% to 200% of their annual salary**, depending on seniority and role complexity. For an organization of 1,470 employees with a 16.12% attrition rate, this translates to **237 departures annually** — a retention challenge with material financial and operational consequences.

1.2 Project Objective

This project was undertaken to answer a core business question faced by HR leadership teams:

> *"Who is leaving, why are they leaving, and what can we do about it?"*

The analysis moves beyond surface-level reporting to identify specific departments, roles, and behavioral patterns that predict attrition — enabling HR and management to make targeted, evidence-based interventions rather than broad, costly initiatives.

1.3 Scope

- **Dataset:** IBM HR Employee Attrition (public dataset)
- **Records analyzed:** 1,470 employees, 35 attributes
- **Tool used:** Microsoft Excel (Pivot Tables, Pivot Charts, Slicers, KPI Cards)
- **Analysis dimensions:** Department, Job Role, Gender, Education Field, Overtime, Age, Monthly Income

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2. Methodology

2.1 Data Cleaning

Prior to analysis, the dataset underwent a structured cleaning process to ensure accuracy and consistency:

Cleaning Step	Action Taken
Duplicate records	Checked and confirmed zero duplicates
Missing values	Verified all 1,470 records were complete
Data type standardization	Ensured numerical fields (Age, Income) were formatted correctly
Categorical consistency	Verified consistent spelling across Department, JobRole, Gender fields
Attrition encoding	Confirmed binary encoding (Yes/No) applied consistently

2.2 Analytical Approach

The analysis followed a structured business intelligence workflow:

1. **Define business questions** — Identify what decisions the analysis needs to support
2. **Explore the data** — Understand distributions, ranges, and key variables
3. **Segment and aggregate** — Use Pivot Tables to analyze attrition across dimensions
4. **Identify patterns** — Surface correlations between attrition and key variables
5. **Quantify impact** — Express findings in business-relevant metrics
6. **Visualize findings** — Build an interactive dashboard for stakeholder use
7. **Generate recommendations** — Translate findings into actionable business decisions

2.3 Key Metrics Defined

Metric	Definition
Attrition Count	Number of employees who left (Attrition = Yes)
Attrition Rate	(Attrition Count ÷ Total Employees) × 100
Average Age by Attrition	Mean age of employees who left vs. stayed
Average Income by Attrition	Mean monthly income of employees who left vs. stayed
Overtime Attrition Rate	Attrition rate among overtime vs. non-overtime employees

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3. Findings

3.1 Overall Attrition

Metric	Value
Total Employees	1,470
Employees Left	237
Employees Retained	1,233
Attrition Rate	**16.12%**
Industry Benchmark	13.2%
Variance from Benchmark	**+2.92 percentage points**

The 16.12% attrition rate exceeds the industry average by nearly 3 percentage points. This is not a marginal deviation — it represents approximately **43 additional departures per year** compared to industry norms, each carrying significant replacement costs.

3.2 Attrition by Department

Department	Total Employees	Employees Left	Attrition Rate
Sales	446	92	**20.63%**
Human Resources	63	12	**19.05%**

Department	Total Employees	Employees Left	Attrition Rate
Research & Development	961	133	**13.84%**

Key finding: Sales and Human Resources both exceed the company average of 16.12% by department rate. Research & Development, despite accounting for the highest volume of attrition in absolute numbers (133 employees), has the lowest attrition *rate* (13.84%) — reflecting its large employee base rather than a disproportionate retention problem.

Business implication: Retention resources should be prioritized toward Sales and HR, where the *rate* — not just the count — signals a structural problem.

3.3 Attrition by Job Role

Job Role	Attrition Rate
Sales Representative	**39.76%**
Sales Executive	17.48%
Research Scientist	16.10%
Laboratory Technician	23.94%
Human Resources	23.08%
Manager	4.90%
Manufacturing Director	6.90%
Research Director	2.50%
Healthcare Representative	6.87%

Key finding: Sales Representatives have an attrition rate of 39.76% — more than double the company average. Nearly 1 in 3 Sales Representatives leaves the organization annually. Laboratory Technicians (23.94%) and HR professionals (23.08%) also show significantly elevated rates.

Business implication: Role-specific interventions are required for Sales Representatives, Laboratory Technicians, and HR staff. A blanket retention policy will be insufficient — each role likely has distinct drivers requiring targeted responses.

3.4 Overtime Impact on Attrition

Overtime Status	Employees	Attrition Count	Attrition Rate
Works Overtime	416	127	**30.53%**
No Overtime	1,054	110	**10.44%**

Key finding: Employees who work overtime leave at nearly **3× the rate** of those who do not. This is the single strongest predictor of attrition identified in this analysis.

Business implication: Overtime is not a neutral productivity tool — it carries a measurable retention cost. Departments with high overtime prevalence should be prioritized for workload review and resource allocation.

3.5 Age and Attrition

Attrition Status	Average Age
Left the organization	**33.61 years**
Stayed with the organization	**37.56 years**
Difference	**3.95 years**

Key finding: Employees who leave are, on average, nearly 4 years younger than those who stay. This indicates a disproportionate loss of early-to-mid career talent — the segment with the highest long-term value potential for the organization.

Business implication: The organization is experiencing an early-career retention problem. Without intervention, this creates a talent pipeline gap that will compound over time as experienced employees retire without adequate successors.

3.6 Income and Attrition

Attrition Status	Average Monthly Income
Left the organization	**₹4,787**
Stayed with the organization	**₹6,833**
Difference	**₹2,046/month**

Key finding: Employees who left earned, on average, ₹2,046 less per month than those who stayed. While income is not the sole driver of attrition, this gap — representing approximately 30% lower compensation — is statistically significant.

Business implication: Compensation competitiveness, particularly for junior and mid-level roles, should be evaluated against market benchmarks. Targeted salary adjustments in high-attrition roles may deliver measurable retention improvements.

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4. Dashboard Design Decisions

Why these charts were chosen

Chart	Type	Business Reason
Attrition Count by Department	Clustered Bar	Compare absolute volume across 3 departments
Department Attrition Rate (%)	Horizontal Bar	Compare rates — prevents large-department bias
Attrition Rate by Job Role	Horizontal Bar	Many categories — horizontal bars are easier to read
Attrition Rate by Overtime	Pie Chart	Binary comparison — clear visual split
Average Age by Attrition	Column Chart	Direct before/after comparison of a single metric
Average Monthly Income by Attrition	Column Chart	Direct before/after comparison of a single metric

Why slicers were chosen over manual filters

Slicers allow non-technical stakeholders — HR managers, department heads, executives — to interact with the data without any Excel knowledge. A hiring manager can filter to "Sales" + "Male" in two clicks and see the complete attrition picture for that segment instantly. This transforms the dashboard from a static report into a decision-support tool.

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5. Recommendations

Priority 1 — Immediate (0–3 months)

Targeted retention program for Sales Representatives With a 39.76% attrition rate, Sales Representatives require immediate intervention. Conduct structured exit interviews with recently departed Sales Reps to identify role-specific drivers. Review commission structure, quota fairness, and career growth visibility in the Sales function.

Priority 2 — Short-term (3–6 months)

Overtime audit and workload redistribution The 3× attrition rate among overtime employees is the clearest signal in the data. Conduct a department-level overtime audit to identify structural causes (understaffing, poor workload distribution, unrealistic targets). Implement overtime caps and pilot flexible work arrangements in high-risk departments.

Priority 3 — Medium-term (6–12 months)

Early-career compensation and development strategy The consistent pattern of younger, lower-paid employees leaving requires a structural response. Benchmark junior and mid-level salaries against the market. Introduce transparent promotion criteria, mentoring programs, and structured learning paths to improve early-career retention.

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6. Limitations and Assumptions

Limitation	Impact
Dataset is cross-sectional (single point in time)	Cannot track individual employee journeys over time
No exit interview data available	Cannot confirm the self-reported reasons for leaving
Income data is monthly, not total compensation	May underrepresent true compensation including bonuses
Dataset is public and synthetic	Findings are directionally valid but should be validated against live organizational data

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7. Conclusion

This analysis demonstrates that IBM's employee attrition challenge is not uniform — it is concentrated in specific roles, departments, and behavioral contexts that are identifiable and, therefore, addressable.

The data tells a clear story: **Sales Representatives are leaving at nearly 40%, overtime doubles attrition risk, and the organization is losing its youngest and lowest-paid employees at a disproportionate rate.** These are not coincidences — they are patterns that respond to targeted business decisions.

The interactive dashboard built for this project enables HR leadership to move from intuition-based retention efforts to evidence-based interventions — filtering by any combination of department, gender, education, and job role to understand the specific attrition profile of any employee segment in real time.

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8. Project Details

Detail	Information
Analyst	Sowmya Sanikommu
Tool Used	Microsoft Excel 2021
Dataset	IBM HR Employee Attrition (Public Dataset)
Records Analyzed	1,470 employees, 35 attributes
Project Type	Self-directed Data Analytics Portfolio Project
Year	2026
Contact	[Your LinkedIn / Email]

This report was prepared as part of a Data Analytics Portfolio project simulating a real-world HR analytics engagement.